Year 11 Set 2 · Higher · grade 5–6 · Lesson 6

Probability tree diagrams

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Today I will learn to...

Goals

1. Use tree diagrams for independent events

2. Use tree diagrams without replacement

Do now: retrieval practice

4 mins

1. 0.3×0.4

   0.12

2. $1 - 0.65$

   0.35

3. $2/5 \times 3/4$

   3/10

4. $3/7 + 2/7$

   5/7

Objective 1

Use tree diagrams for independent events

Key idea

Learn

!

Multiply along the branches. Add the results for different routes. 'At least one' = 1 − P(none).

Objective 1: Use tree diagrams for independent events

I do: example 1

I do

P(rain) = 0.3 each day. Find P(rain on both of 2 days).

1. Rain on day 1 AND rain on day 2
2. AND → multiply along the branches
3. 0.3×0.3
4. $= 0.09$

I do: example 2

I do

Same days. Find P(rain on exactly one day).

1. Two ways: rain then no rain, or no rain then rain
2. Rain, no rain: $0.3 \times 0.7 = 0.21$
3. No rain, rain: $0.7 \times 0.3 = 0.21$
4. OR → add: $0.21 + 0.21 = 0.42$

I do: example 3

I do

Same days. Find P(rain on at least one day).

1. At least one = 1 − P(no rain on both days)
2. P(no rain on both) = 0.7 × 0.7 = 0.49
3. 1 − 0.49
4. = 0.51

Mini whiteboards

We do

1. 3 red, 2 blue, with replacement. P(both red)?

Answer: 9/25

2. P(heads) = 0.5, two flips. P(two heads)?

Answer: 0.25

3. P(rain) = 0.3 each day. P(no rain on two days)?

Answer: 0.49

4. P(pass) = 0.8 per test, two tests. P(pass exactly one)?

Answer: 0.32

You do: choose your challenge

You do

Bronze

P(A) = 0.4 each time, independently. Find P(A happens twice).

Silver

P(hit) = 0.7 per shot. Find P(miss both of two shots).

Gold

P(hit) = 0.7 per shot. Find P(exactly one hit in two shots).

Platinum

P(hit) = 0.7 per shot. Find P(at least one hit in three shots).

You do: answers

Check

Bronze	0.16
Silver	0.09
Gold	0.42
Platinum	$1 - 0.3^3 = 0.973$

If you finish early

Extra

Bronze

P(win) = 0.6 each game. Find P(winning two games).

0.36

Silver

P(red light) = 0.4 at each of two lights. Find P(no red lights).

0.36

Gold

P(score) = 0.75 per penalty. Find P(exactly one goal from two penalties).

$2 \times 0.75 \times 0.25 = 0.375$

Platinum

P(late) = 0.1 each day. Find P(late at least once in 3 days).

$1 - 0.9^3 = 0.271$

Exam question 1

3 marks

P(Ali's bus is late) = 0.2 each day. Find P(late on exactly one of two days).

Space for working

Mark scheme: $2 \times 0.2 \times 0.8 = 0.32$

Exam question 2

2 marks

Find P(late on at least one of the two days).

Space for working

Mark scheme: $1 - 0.8^2 = 0.36$

Objective 1 achieved

Use tree diagrams for independent events



Objective 2

Use tree diagrams without replacement

✓ Achieved: Use tree diagrams for independent events

Key idea

Learn

!

On the second branches the total goes down by 1, and so does the count of the colour already taken.

Objective 2: Use tree diagrams without replacement

I do: example 1

I do

5 red and 3 green. Two taken. P(both red)?

1. First red: 5/8
2. Second red (one red gone): 4/7
3. Multiply: 5/8 × 4/7 = 20/56
4. Simplify: = 5/14

I do: example 2

I do

Same bag. P(same colour)?

1. Same colour = both red OR both green
2. Both red: $5/8 \times 4/7 = 20/56$
3. Both green: $3/8 \times 2/7 = 6/56$
4. Add: $26/56 = 13/28$

I do: example 3

I do

Same bag. P(different colours)?

1. Different colours is the opposite of the same colour
2. P(same colour) = 13/28
3. 1 − 13/28
4. = 15/28

Mini whiteboards

We do

1. 4 blue, 6 white, two taken. P(both blue)?

Answer: 2/15

2. 5 red, 3 green, two taken. P(both green)?

Answer: 3/28

3. Same bag. P(both red)?

Answer: 5/14

4. Same bag. P(one of each)?

Answer: 15/28

You do: choose your challenge

You do

Bronze

3 red and 2 blue, two taken without replacement. Find P(both blue).

Silver

Same bag. Find P(both red).

Gold

Same bag. Find P(different colours).

Platinum

A bag has n sweets, 6 of them orange. P(taking two orange without replacement) = 1/3. Find n.

You do: answers

Check

Bronze	1/10
Silver	3/10
Gold	3/5
Platinum	$30/(n(n-1)) = 1/3$, $n^2 - n - 90 = 0$, $n = 10$

If you finish early

Extra

Bronze

A drawer has 6 black and 4 white socks. Two are taken. Find P(both black).

30/90 = 1/3

Silver

Same drawer. Find P(both white).

12/90 = 2/15

Gold

Same drawer. Find P(a matching pair).

1/3 + 2/15 = 7/15

Platinum

A bag has 3 red and n blue balls. Two are taken. P(both red) = 1/15. Find n.

6/((n + 3)(n + 2)) = 1/15, (n + 3)(n + 2) = 90, n = 7

Exam question 1

4 marks

A bag has 4 red and 3 yellow counters. Two are taken without replacement. Find P(one of each colour).

Space for working

Mark scheme: $2 \times 4/7 \times 3/6 = 4/7$

Objective 2 achieved

Use tree diagrams without replacement



Real life

Apply

P(passing a driving test first time) = 0.7. If you fail, P(passing the second time) = 0.8. Find P(passing within two attempts).

$0.7 + 0.3 \times 0.8 = 0.94$

Extension: stretch yourself

Stretch

1. A box has 10 sweets, 3 orange. Sam eats two at random. Find P(at least one orange).

$1 - 7/10 \times 6/9 = 8/15$

2. Three counters are taken without replacement from 4 red and 2 blue. Find P(all red).

$4/6 \times 3/5 \times 2/4 = 1/5$

Exit ticket: exam questions

Exit

1. P(rain) = 0.4 each day. Find P(no rain on either of two days). **(2 marks)**

2. Find P(rain on exactly one of the two days). **(2 marks)**

3. A bag has 5 red and 4 blue counters. Two are taken without replacement. Find P(both the same colour). **(4 marks)**

Exit ticket: mark scheme

Mark it

Q1 0.36 [2]

Q2 0.48 [2]

Q3 20/72 + 12/72 = 4/9 [4]

Total: _____ / 8

Today I achieved

- ✔ **Use tree diagrams for independent events**
- ✔ **Use tree diagrams without replacement**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. P(win) = 0.3 for each game. Find P(winning both of two games). [2]

2. Find P(winning at least one of the two games). [2]

3. A bag has 6 green and 4 yellow counters. Two are taken without replacement. Find P(both green). [3]

4. Same bag. Find P(one of each colour). [3]

5. P(pass theory) = 0.8. If you pass theory, P(pass practical) = 0.65. If you fail theory, P(pass practical) = 0.4. Find P(pass practical). [3]

Extra practice: Maths Genie → Edexcel question bank → Probability Trees